Classical Optical Interferometry

Mach–Zehnder and Michelson Interferometers

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Abstract

This experiment explored the physics and practical implementation of classical optical interferometry using a Mach–Zehnder interferometer and a Michelson interferometer. The experiments investigated interference arising from optical path differences, wavefront curvature, polarization labeling, and coherence properties of a HeNe laser source. We included ray-tracing optics and wave optics approaches for analyzing the experiment, including Gaussian beam propagation, paraxial approximation, and matrix-based ray tracing methods. Observations of fringe formation, interference visibility, and coherence effects were used to connect theoretical predictions with experimental measurements.

1 Introduction

Interferometry is one of the most powerful techniques in optics, enabling the measurement of phase, displacement, wavelength, and coherence properties of light with extremely high precision. The fundamental principle underlying interferometry is the superposition of coherent electromagnetic waves. When two optical fields overlap, their electric field amplitudes add, producing an intensity pattern that depends on the phase difference between them. This phase difference can arise from differences in optical path length, refractive index, wavefront curvature, or polarization.

In this laboratory, two canonical interferometric systems were constructed and studied: the Mach–Zehnder interferometer and the Michelson interferometer. Both are amplitude-splitting interferometers, meaning that an incoming optical field is divided into two spatially separate paths using a beamsplitter. The two beams are later recombined, allowing interference to occur. Despite their conceptual similarity, the two geometries emphasize different physical principles. The Mach–Zehnder interferometer provides a clear visualization of interference between independent optical paths and enables manipulation of wavefront geometry and polarization. The Michelson interferometer, in contrast, folds the optical paths back onto themselves and is particularly sensitive to changes in path length, forming the basis of precision metrology and gravitational-wave detection.

The experiments were conducted using a coherent HeNe laser at a wavelength of 633 nm. In this classical regime, interference arises from continuous electromagnetic waves. However,

the same physical behavior persists in the quantum limit, where single photons pass through the interferometer one at a time and interference emerges from accumulated detection events. Thus, the classical optical interferometer serves as a direct analogue to quantum interference phenomena and provides insight into foundational concepts such as which-path information and coherence.

This report develops a detailed theoretical framework for understanding interferometry in the context of the performed experiments. The discussion includes ray propagation using ABCD matrices, the paraxial approximation, Gaussian beam optics, polarization effects, and the formation of linear and spherical interference fringes. These theoretical elements are then connected to the construction, alignment, and operation of both interferometers in the laboratory.

2 Theory

2.1 Interference and Optical Path Difference

Interference arises when two coherent electromagnetic waves overlap in space. The total electric field at a point is given by the superposition of the individual fields,

$$E(\mathbf{r}, t) = E_1(\mathbf{r}, t) + E_2(\mathbf{r}, t). \tag{1}$$

The measurable quantity is the intensity,

$$I \propto \langle |E|^2 \rangle. \tag{2}$$

Expanding,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\Delta\phi), \tag{3}$$

where $\Delta\phi$ is the phase difference between the two waves.

This phase difference is determined by the optical path difference (OPD),

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta L_{\text{opt}}, \tag{4}$$

where

$$\Delta L_{\text{opt}} = n_1 L_1 - n_2 L_2, \tag{5}$$

with n the refractive index and L the geometric path length.

In interferometers, beams traverse different optical paths before recombination. Constructive interference occurs when

$$\Delta L_{\text{opt}} = m\lambda, \tag{6}$$

while destructive interference occurs for

$$\Delta L_{\text{opt}} = \left(m + \frac{1}{2}\right)\lambda. \tag{7}$$

Thus, even extremely small changes in path length produce measurable intensity changes. This sensitivity is the fundamental reason interferometers are used in precision metrology.

2.2 Paraxial Approximation

In most laboratory optical systems, beams propagate at small angles relative to the optical axis. Under these conditions, the paraxial approximation can be applied.[1] The electric field envelope satisfies the paraxial wave equation,

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} - 2ik\frac{\partial U}{\partial z} = 0, \tag{8}$$

where $U(x, y, z)$ is the slowly varying amplitude and $k = 2\pi/\lambda$.

This approximation assumes:

- small propagation angles
- slowly varying transverse beam profile
- negligible back-reflection

For the interferometers constructed in this experiment, the paraxial regime is valid because the beam divergence is small and optical elements are aligned near normal incidence. This allows beams to be treated as quasi-planar wavefronts and enables Gaussian beam solutions.

2.3 Gaussian Beam Optics

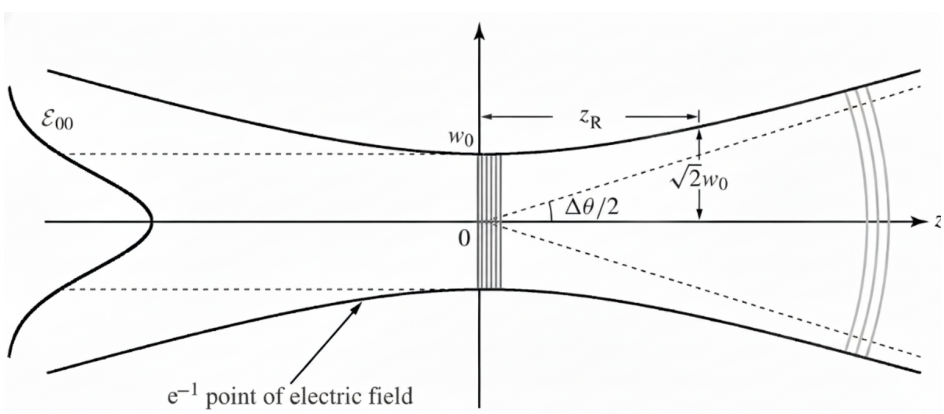


Figure 1: The profile of a paraxial beam with gaussian distribution of energy density

The HeNe laser output is well described by a Gaussian beam[1]. The electric field amplitude is given by

$$E(r, z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{r^2}{w^2(z)}\right) \exp\left[-ikz - i\frac{kr^2}{2R(z)} + i\zeta(z)\right]. \tag{9}$$

Key parameters include:

Beam waist: Which is the minimum diameter of the gaussian beam.

$$w_0 \tag{10}$$

Beam radius: Which is the radius of the gaussian beam from the optical axis at any given plane z.

$$w(z) = w_0\sqrt{1 + \left(\frac{z}{z_R}\right)^2} \tag{11}$$

Rayleigh range: The Rayleigh range Z_r defines the distance from the waist along which the cross-sectional area of the beam approximately doubles. It is a measure of how long the beam stays relatively collimated before diffraction dominates.

$$z_R = \frac{\pi w_0^2}{\lambda} \tag{12}$$

Where λ is the wavelength of light. Within the Rayleigh range, the wavefront is nearly planar; beyond it, the wavefront curvature becomes significant. In interferometry, the Rayleigh range helps determine the region over which the beams overlap efficiently and produce high-contrast fringes.

Radius of curvature: The radius of curvature R(z) describes the curvature of the Gaussian beam's wavefronts at a distance z from the waist. A smaller radius corresponds to a more curved (spherical) wavefront, while $R \to \infty$ corresponds to a planar wavefront.

$$R(z) = z\left[1 + \left(\frac{z_R}{z}\right)^2\right] \tag{13}$$

This curvature is important in generating different fringe patterns. For example, overlapping beams with different R(z) values produces spherical (circular) fringes, whereas nearly planar wavefronts produce linear fringes.

Gouy phase: The Gouy phase shift ζ(z) represents an additional phase that a Gaussian beam acquires as it propagates through the focus. Unlike a plane wave, which accumulates phase linearly with kz, the Gaussian beam acquires an extra phase contribution.

$$\zeta(z) = \tan^{-1}\left(\frac{z}{z_R}\right) \tag{14}$$

This phase shift affects the interference pattern when two beams with different waists or focusing positions are overlapped.

The gaussian beam equation therefore describes the width of the beam at a particular horizontal plane z, the spatial distribution of E(r,z) accross a wavefront described by $exp(r^2/w^2(z))$ term, the wavefront propogation term given by $exp(-ikz)$ and the gouy phase term.

2.4 Ray Propagation and ABCD Matrices

In the paraxial regime, optical systems can be modeled using ray transfer matrices[2]. A ray is described by

$$\begin{pmatrix} x \\ \theta \end{pmatrix}, \tag{15}$$

where x is transverse position and θ is propagation angle.

Propagation through optical components is described by

$$\begin{pmatrix} x_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x_1 \\ \theta_1 \end{pmatrix}. \tag{16}$$

Important matrices used in this lab include:
Free-space propagation (distance L)

$$\begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \tag{17}$$

Thin lens (focal length f)

$$\begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix} \tag{18}$$

Mirror reflection

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{19}$$

ABCD matrices allow prediction of beam waist evolution, collinearity of beams, and the effect of lenses used to modify wavefront curvature. They are essential for aligning interferometers and ensuring overlapping optical modes.

2.5 Beam Splitters and Phase Relations

A non-polarizing 50:50 beamsplitter divides an incident field into transmitted and reflected components of equal amplitude. The reflected beam typically acquires a phase shift of $\pi/2$ relative to the transmitted beam. Thus, if an input field E_0 enters a beamsplitter, the output amplitudes may be written as

$$E_T = \frac{E_0}{\sqrt{2}}, \qquad E_R = \frac{iE_0}{\sqrt{2}}. \tag{20}$$

When two beams recombine at the second beamsplitter, the total field at each output port is a superposition of transmitted and reflected components from both arms. The resulting intensity depends on the relative phase accumulated along the two paths. This produces complementary outputs: when one port is bright, the other is dark.

In an ideal interferometer with equal path lengths and perfect alignment, all light exits a single output port due to constructive interference, while the second port exhibits destructive interference. Deviations from ideal alignment produce interference fringes, which provide a direct measure of phase variation across the beam profile.

2.6 Mach–Zehnder Interferometer: Physical Principles

The Mach–Zehnder interferometer is an amplitude-splitting interferometer in which an incoming optical beam is divided into two spatially separated paths using a beamsplitter[1]. Each beam propagates independently through free space and reflective optics before being

recombined at a second beamsplitter. Interference occurs at the output ports as a result of phase differences accumulated along the two paths.

The key physical quantity governing the output is the relative phase between the two beams. For a monochromatic field, the phase accumulated along a path of optical length L_{opt} is

$$\phi = \frac{2\pi}{\lambda} L_{\text{opt}}. \tag{21}$$

Thus, even sub-wavelength differences in path length produce observable changes in output intensity. If the two paths are identical, the phase difference is zero and constructive interference occurs at one detector while destructive interference occurs at the other. In practice, however, perfect path equality is impossible, and slight misalignments produce spatially varying interference fringes.

A defining feature of the Mach–Zehnder geometry is that it allows independent manipulation of each beam before recombination. Lenses may be inserted to alter wavefront curvature, polarizers may encode path information, and mirrors can introduce controlled angular misalignment. Because the beams propagate separately, the Mach–Zehnder interferometer provides a direct visualization of how phase, polarization, and wavefront geometry influence interference.

Mach–Zehnder interference using matrix multiplication

Let the 50:50 beam splitter be represented by

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} i & 1 \\ 1 & i \end{pmatrix}, \qquad \text{and input state } |\psi_{\text{in}}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

After the first beam splitter:

$$|\psi_1\rangle = B\,|\psi_{\text{in}}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix}.$$

For equal arm lengths ($\Delta\phi = 0$), propagation does not change the relative phase, so $|\psi_2\rangle = |\psi_1\rangle$. Recombining at the second beam splitter:

$$|\psi_{\text{out}}\rangle = B\,|\psi_2\rangle = \frac{1}{2} \begin{pmatrix} i & 1 \\ 1 & i \end{pmatrix} \begin{pmatrix} i \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

Thus the output intensity is

$$I_{D_0} = 1, \qquad I_{D_1} = 0,$$

showing constructive interference at D_0 and destructive interference at D_1.

2.7 Fringe Formation and Wavefront Geometry

Interference fringes arise from the spatial variation of the phase difference between two coherently overlapping optical beams. The geometry of the observed fringe pattern is therefore

determined by the relative wavefront shapes and propagation directions of the interfering beams. In a Mach–Zehnder interferometer, this geometry can be controlled by adjusting alignment and by inserting lenses into one or both arms, thereby modifying the curvature of the propagating wavefronts.

Interference of two plane waves (linear fringes) Consider two linearly polarized plane waves that intersect at a small angle θ. One wave propagates along the z-axis,

$$E_1(\mathbf{r}, t) = A_1 \cos(kz - \omega t),$$

while the second wave propagates at a small angle θ relative to it,

$$E_2(\mathbf{r}, t) = A_2 \cos(kz \cos\theta + kx \sin\theta - \omega t).$$

The total electric field is the superposition

$$E(\mathbf{r}, t) = E_1 + E_2,$$

and the observed intensity is proportional to the time-averaged square of the field,

$$I \propto \langle E^2 \rangle.$$

Evaluating the cross term gives the interference modulation

$$I(x) = I_0 \left[1 + \cos\left(\frac{2\pi}{\lambda} x \sin\theta \right) \right].$$

The phase difference varies linearly with transverse coordinate x, producing straight, equally spaced fringes. The separation between adjacent fringes is

$$\Delta x = \frac{\lambda}{\sin\theta}.$$

Thus, small angular misalignment between two nearly planar wavefronts produces linear interference fringes. The fringe spacing is determined by the wavelength and the relative angle between the beams.

Interference of a plane wave and a spherical wave When one beam has a planar wavefront and the other has a spherical wavefront of radius of curvature R, the electric fields may be written as

$$E_1 \propto \cos(kz - \omega t),$$

$$E_2 \propto \cos\left(kz + \frac{k(x^2 + y^2)}{2R} - \omega t \right).$$

The spatially varying phase difference is therefore

$$\Delta\phi(r) = \frac{kr^2}{2R}, \qquad r^2 = x^2 + y^2.$$

The interference term becomes

$$I(r) \propto 1 + \cos\left(\frac{kr^2}{2R}\right),$$

which produces concentric circular fringes centered on the optical axis. The bright fringe radii occur where the phase difference equals integer multiples of 2π,

$$r_m = \sqrt{2Rm\lambda}, \qquad m = 0, 1, 2, \ldots$$

This pattern is commonly observed when a lens is introduced into one arm of the interferometer, creating curvature in one of the wavefronts.

Interference of two spherical wavefronts If both beams have spherical wavefronts with radii R_1 and R_2, their phases can be written as

$$\phi_1 = \frac{kr^2}{2R_1}, \qquad \phi_2 = \frac{kr^2}{2R_2}.$$

The phase difference becomes

$$\Delta\phi(r) = \frac{kr^2}{2}\left(\frac{1}{R_1} - \frac{1}{R_2}\right).$$

The resulting intensity modulation is

$$I(r) \propto 1 + \cos\left[\frac{kr^2}{2}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)\right],$$

again producing circular fringes. The spacing and curvature of the rings depend on the difference in radii of curvature.

1) If $R_1 = R_2$, the phase difference becomes constant and no fringes are observed, indicating identical wavefront curvature.

2)If $R_1, R_2 = \infty$, the beams are significantly diverged and the wavesfronts are almost are planar and there are no fringes. But if both the beams are not perfectly overlapping then there will still be planar fringes.

Interference of two plane waves with perfect alignment If both beams are planar and perfectly parallel, their phase difference is constant across the transverse plane:

$$\Delta\phi = \text{constant}.$$

The intensity becomes uniform,

$$I = \text{constant},$$

and no fringes are visible. This corresponds to the ideal alignment condition in an interferometer, where all optical power exits a single port.

In the Mach–Zehnder interferometer, these observations are used as alignment diagnostics. Perfectly parallel and identical wavefronts produce no spatial fringes and direct all

optical power to detector D_0, while detector D_1 remains dark. Slight angular misalignment produces linear fringes, and curvature differences introduced by lenses produce spherical (ring) fringes where in D_0 produces bright fringe at the center and D_1 produces dark fringe at the center giving mutually opposite fringe patterns. Thus, the observed fringe patterns provide direct information about beam alignment, propagation geometry, and wavefront curvature in the interferometer.

2.8 Polarization and Which-Path Information, and Jones calculus

Jones calculus description of polarization Jones calculus represents polarization states as vectors and optical elements as matrices[3]. A general polarization state is written as

$$\mathbf{E} = \begin{pmatrix} E_x \\ E_y \end{pmatrix}.$$

An ideal linear polarizer oriented along the horizontal axis is described by the matrix

$$P_H = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

while a vertical polarizer is

$$P_V = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

If an arbitrary field passes through a horizontal polarizer,

$$\mathbf{E}_{\text{out}} = P_H \mathbf{E} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} E_x \\ 0 \end{pmatrix},$$

showing that the polarizer projects the electric field onto a single polarization axis and removes the orthogonal component.

A polarizer oriented at an angle θ to the horizontal is described by

$$P(\theta) = \begin{pmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{pmatrix}.$$

Polarization using which–path information

Interference in a Mach–Zehnder interferometer requires the two beams to be physically indistinguishable when they recombine. If some property of the light allows us to determine which path it took, the interference term disappears. One important way to introduce such distinguishability is through polarization. A linear polarizer works by passing light in a particular polarization axis (for example horizontal axis for horizontal polarizer) and blocking light in all other axes, thus reducing the intensity. In this case, the electric field must be treated as a vector quantity rather than a scalar amplitude[3],

$$\mathbf{E} = E_x\hat{x} + E_y\hat{y}.$$

Suppose the two arms of the interferometer are labeled with orthogonal polarizations. One arm is horizontally polarized and the other vertically polarized:

$$\mathbf{E}_H = \begin{pmatrix} E_0 \\ 0 \end{pmatrix}, \qquad \mathbf{E}_V = \begin{pmatrix} 0 \\ E_0 \end{pmatrix}.$$

These two polarization states are orthogonal,

$$\mathbf{E}_H \cdot \mathbf{E}_V = 0,$$

so when the beams recombine, the total intensity contains no cross term. The observed intensity is simply the sum of the individual intensities,

$$I = |\mathbf{E}_H|^2 + |\mathbf{E}_V|^2,$$

and no interference fringes appear. Physically, the polarization carries which–path information, making the two beams distinguishable.

However, placing a polarizer after recombination projects both beams onto a common polarization axis. The fields then become indistinguishable again, and interference is restored. This is a classical analogue of the quantum eraser concept, demonstrating how information about a system can determine whether interference occurs.

The transmitted intensity through a polarizer at angle θ follows Malus' law[4],

$$I = I_0 \cos^2 \theta. \tag{22}$$

Thus rotating the polarizer modulates the recovered fringe visibility.

2.9 Michelson Interferometer: Operating Principles

The Michelson interferometer differs from the Mach–Zehnder in that a single beamsplitter both divides and recombines the beam[5]. Light travels along two perpendicular arms, reflects from mirrors, and returns to the beamsplitter where interference occurs.

Because each beam retraces its path, the Michelson geometry is extremely sensitive to changes in path length. A displacement of one mirror by ΔL produces a round-trip path difference of $2\Delta L$, leading to a phase shift

$$\Delta\phi = \frac{4\pi}{\lambda}\Delta L. \tag{23}$$

As a mirror moves, bright and dark fringes sweep across the observation plane. Counting these fringes provides a direct measurement of displacement or wavelength. This sensitivity is the reason Michelson interferometers are widely used in precision measurement and spectroscopy.

2.10 Fringe Visibility and Coherence

Interference requires temporal and spatial coherence. Temporal coherence is related to the spectral bandwidth of the source, while spatial coherence describes the uniformity of phase across the beam profile[6].

The visibility of interference fringes is defined as

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}. \tag{24}$$

Visibility quantifies the strength of contrast between the interference fringes. It can decrease due to 1) a mismatch between radius of curvatures of the beams, 2)Bad Geometric overlap between beams 3) Spatial phase difference creating an angular mismatch between wave-fronts, 4)Polarization mismatch between the beams. Also the path difference between the two arms increases, the phase relationship between the beams becomes less well defined, and fringe visibility decreases. The variation of visibility with the path difference is given by-

Visibility Curve:

$$V(\Delta L) = \exp\left[-\left(\frac{\pi \Delta L}{2L_c}\right)^2\right] \tag{25}$$

Where, the characteristic distance over which interference remains observable is the coherence length. The coherence length can thus can be approximated using-

$$Visibility_{L_c} = 1/e * (Visibility_{max}) \tag{26}$$

He-Ne lasers typically exhibit coherence lengths on the order of tens of centimeters to meters, depending on spectral linewidth. In the Michelson interferometer, translating one mirror changes the path difference and allows direct measurement of fringe visibility as a function of arm length. By fitting this dependence, the coherence length of the source can be estimated.

3 Procedure

3.1 Mach–Zehnder Interferometer

The Mach–Zehnder interferometer was assembled following the steps to build the setup shown in figure 2. First, the He-Ne laser was securely mounted, and beam steering mirrors were placed to guide the beam onto the first 50:50 non-polarizing beam-splitter. This beam-splitter divided the input beam into two arms, each reflecting off a silver mirror mounted on kinematic mounts, before recombining at the second beam-splitter. The path from the first to the second beam splitter was designed to be around 39 $\pm$1 cm for both the arms.

The interferometer was initially aligned by observing the direct output beam (D0) projected onto a wall. The interference fringes were visible, indicating that the optical paths were nearly matched. To enhance the visibility of the fringe pattern, a 100 mm focal length converging lens was placed in the path to D0 10$\pm$1 cm downstream from the BS2. This was done in order to enlarge the beam on the wall. Adjustments to the tilt and rotation of the mirrors and beamsplitters were iteratively performed until a clear, centered interference pattern was obtained.

Once aligned, the distance between the nearest fringe pair was measured. Then two Si photo-detectors were placed at the output ports, connected to a CRO with impedance

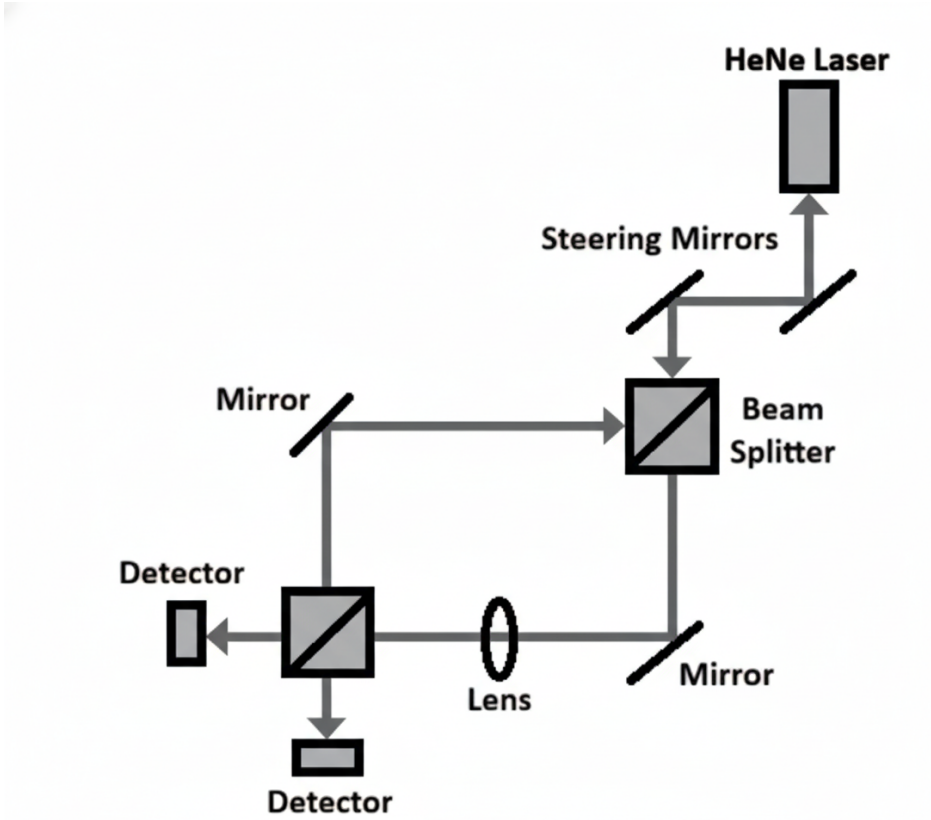


Figure 2: The setup of Mach-Zehnder Interferometer experiment

matching.The impedance used was 10kohm. The optical power at both detectors was measured to quantify the visibility and contrast of the interference using the V_{max} and V_{min} of the observed sinusoids(using equation 24).

Additional converging lenses with focal length 50mm were inserted to alter the wavefronts of one or both beams, converting plane waves into spherical waves by collimating the beam so that the radius of curvature R becomes much less than that of planar waves (ideally R = ∞ relative to λ for planar waves). On the left-arm the lens was placed at a distance of 5cm downstream from the mirror, whereas on the right-arm it was 10cm downstream. This was done to maximize the path-difference between the interfering spherical-spherical waves. The resulting fringe patterns were recorded for each configuration namely - 1) One Spherical wave interfering with Planar wave 2) Spherical wave interfering with Spherical wave and 3) Planar wave interacting with Planar wave.

3.2 Polarization and Which-Path Information with Spherical Waves

Polarizers were introduced into both arms of the interferometer (with the spherical-wave–generating lenses left in place) in order to investigate the role of which-path information in the formation of interference fringes. The aim was to demonstrate that interference depends on the indistinguishability of the two optical paths and that polarization can be used to control this distinguishability.

Linear polarizers were placed in each arm approximately 10 cm downstream of BS1. The left arm was fitted with a horizontal polarizer, while the right arm was fitted with a vertical polarizer. From section 2.8, we see that after transmission through the polarizers, the electric fields in the two arms can be written as vector fields

$$\mathbf{E}_L = E_0 \cos(\omega t)\,\hat{x}, \qquad \mathbf{E}_R = E_0 \cos(\omega t + \phi)\,\hat{y},$$

where $\hat{x}$ and $\hat{y}$ represent mutually orthogonal polarization directions and ϕ is the phase difference accumulated between the two paths.

Because the polarization vectors are orthogonal,

$$\hat{x} \cdot \hat{y} = 0,$$

the interference term vanishes when the intensity is calculated:

$$I \propto \langle |\mathbf{E}_L + \mathbf{E}_R|^2 \rangle = I_L + I_R + 2\langle \mathbf{E}_L \cdot \mathbf{E}_R \rangle = I_L + I_R.$$

Each polarizer transmits only one component of the incoming light, so the intensity in each arm is reduced to approximately half of the original value.

To restore interference, a third polarizer was placed at the output port D0, mounted on a rotation stage approximately 15 cm downstream of BS2. This polarizer was rotated to an angle of 45° and 135° with respect to the horizontal axis. The action of this polarizer is to project both orthogonally polarized beams onto a common polarization direction.

Using Malus' law,

$$I = I_0 \cos^2 \theta,$$

each beam emerging from the interferometer and passing through the 45° polarizer transmits a fraction

$$I = \frac{I_0}{2} \cos^2(45^\circ) = \frac{I_0}{4}.$$

Thus the intensities from both arms become equal and aligned along the same polarization axis.

In vector form, the transmitted electric fields after the 45° polarizer are

$$\mathbf{E}'_L = \frac{E_0}{\sqrt{2}} \cos(\omega t)\,\hat{d}, \qquad \mathbf{E}'_R = \frac{E_0}{\sqrt{2}} \cos(\omega t + \phi)\,\hat{d},$$

where $\hat{d}$ is the unit vector along the diagonal polarization direction. Since both fields now share the same polarization,

$$\hat{d} \cdot \hat{d} = 1,$$

the interference term reappears:

$$I \propto I_L + I_R + 2\sqrt{I_L I_R} \cos \phi.$$

Therefore for the output beam, the total intensity is reduced to roughly one quarter of the original beam intensity from each arm.

Polarizer at 135° and central dark fringe If the output polarizer is rotated to 135°, both beams are again projected onto a common polarization axis, restoring in-distinguishability and therefore interference. The transmitted intensity from each arm is

$$I = \frac{I_0}{2}\cos^2(45^\circ) = \frac{I_0}{4},$$

so interference is recovered with reduced overall intensity, just as in the 45° case.

However, the projection axis at 135° w.r.t to the original propagation axis, introduces an effective relative phase inversion between the two projected field components. As a result, the interference condition at the beam center switches from constructive to destructive:

$$I \propto |E_1 - E_2|^2 ,$$

producing a dark central fringe.

The fringe spacing and geometry should remain identical to the 45° case because they depend only on wavelength and beam alignment, not on the polarization projection angle. Only the phase of the superposed fields changes, causing the central maximum to become a minimum.

Finally, the second beam-splitter was replaced by a plane mirror such that the mirror sends back the light from the left-arm into the right arm and from the right-arm into the left. Thus, the both the arms interfere with each other and the interfered beams are again incident on the first beam-splitter to interfere again. Their fringe pattern is then observed.

3.3 Michelson Interferometer

The interferometer was designed such that the path length of one arm could be easily adjusted. A schematic of the Michelson interferometer is shown in Fig. 3. Following the laser source, two beam-steering mirrors are used to accurately align the beam trajectory and center the beam on the optical components. A non-polarizing 50:50 beam splitter is placed immediately after the steering mirrors, dividing the incident beam into two perpendicular arms, each with an initial path length of 20 ± 1 cm.

Both arms terminate at mirrors with approximately 99% reflectivity. The transmitted arm is directed toward a mirror mounted on a piezoelectric transducer (PZT). The PZT is driven sinusoidally by a function generator at a frequency of 200 Hz, producing small, periodic displacements of the mirror and thereby alternatingly modulating the optical path length of this arm. This modulation introduces a time-dependent phase shift between the two arms, resulting in an oscillatory interference signal at the output of the interferometer. The reflected arm is directed toward a second 99% reflective mirror, which reflects the beam back to the beam splitter.

Upon recombination at the beam splitter, the returning beams interfere and are directed toward the photo-detector. In addition to fine path-length modulation using the PZT, the mirror located opposite the detector was displaced by $\pm 2.5cm, \pm 5.0cm, and \pm 7.5cm$, with a positional uncertainty of ± 0.1 cm. By adjusting the mirror position, the effects of significant optical path length difference on the interference visibility can be investigated.

One important subtlety to note here is that the mirrors are given a small deliberate mismatch angle such that the beams don't interfere with themselves (as seen in fig. 3) and

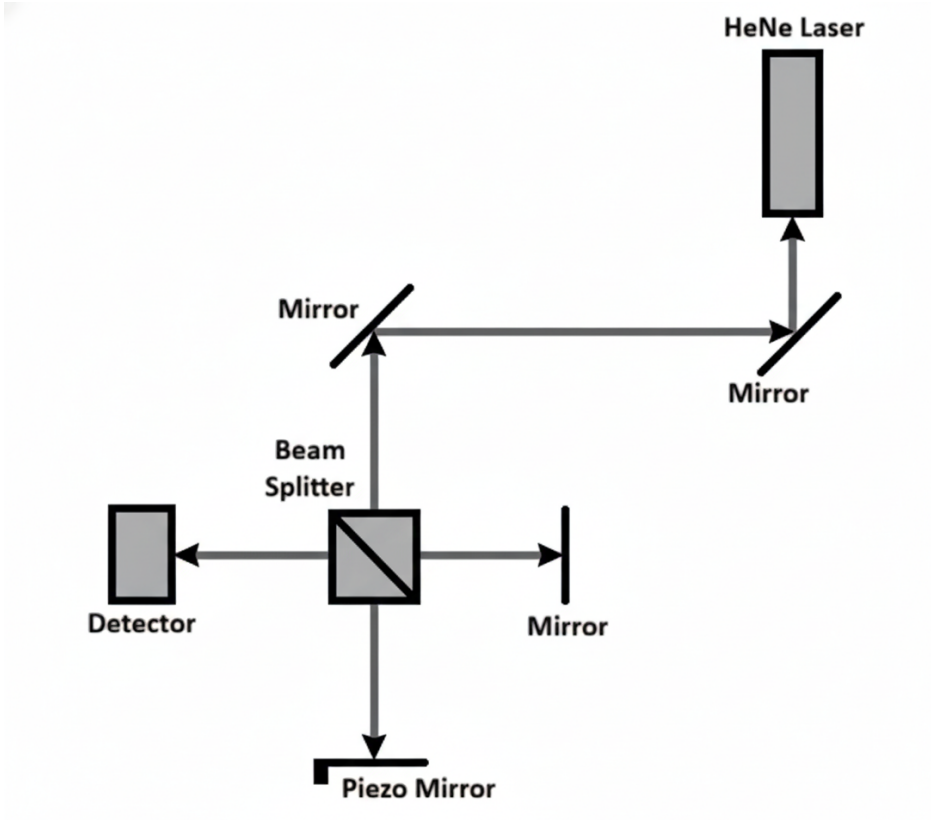


Figure 3: The setup of Michelson Interferometer experiment

only required interference is observed. This although reduces the fringe-contrast and thus lower visibility. We also vary this misalignment angle to 0.286, 0.142, 0.429 degrees and repeat the experiment. We plot the visibility vs the distance and extrapolate the coherence length for each of them using equation 26.

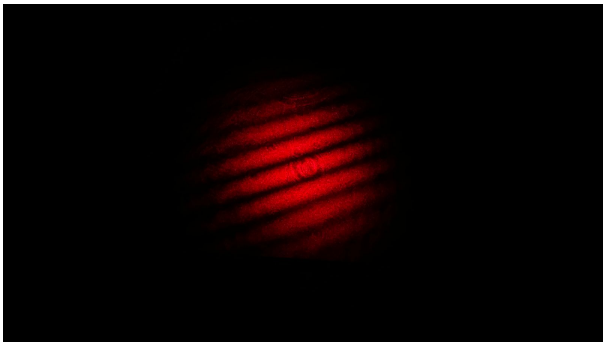
4 Results

4.1 Mach-Zehnder Interferometer

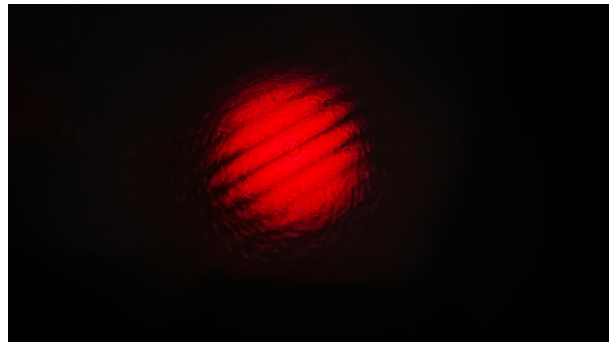
In the Mach–Zehnder interferometer, interference patterns were observed at both output ports. With plane wave-fronts, straight fringes were visible, indicating slight angular misalignment between the two beams (refer fig 4). The pattern at D1 hs a dark fringe in the center whereas at D0 it has a bright fringe at the center because D0 has constructive interference and D1 has destructive interference ideally if the spatial phase diffrence is zero, as we predicted in section 2.6. The distance between the fringe pairs was 1.28 cm±0.1. We calculate the angle between the wavefronts to be-

$$\theta = \sin^{-1}\left(\frac{\lambda}{d}\right) = \sin^{-1}\left(\frac{633nm}{1.28cm}\right) = 0.012rad \tag{27}$$

The photo-detector output can be seen in fig 5. The measured signal in D0 and D1 has a



(a) Constructive interference at detector D0 (bright center).



(b) Destructive interference at detector D1 (dark center).

Figure 4: Fringe patterns observed due to interference of two planar waves in the Mach–Zehnder interferometer at the two detectors.

visible Π radian phase because of opposite fringe pattern due to constructive and destructive interference. The maximum and minimum voltage values for both the detectors are noted in the Table 1 along with their visibilities. The Visibility was calculated from these values using equation 24 and are found to be nearly equal for both the detectors. This confirms that both outputs of the interference translate the same phase difference and intensity parameters.

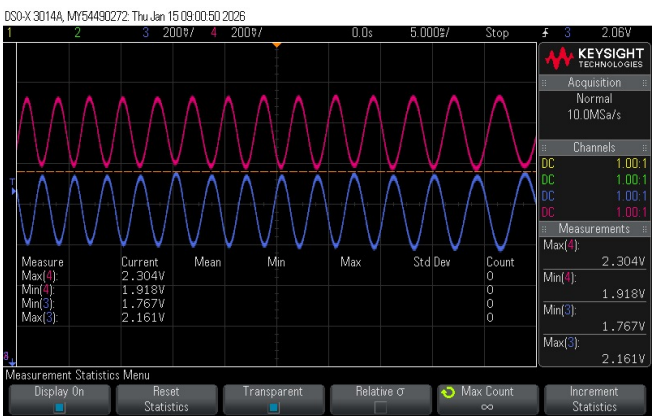


Figure 5: The signal output from detectors D0 and D1 with their maximum and minimum values.

	D0	D1
$V_{\max}$ (V)	2.304	2.161
$V_{\min}$ (V)	1.918	1.767
Visibility	0.091	0.100

Table 1: Measured maximum and minimum detector voltages and calculated fringe visibility for detectors D0 and D1.

In accordance with the theory discussed in section 2.7, we observe to Maximum contrast measured between the bright and dark ports was approximately 9.2% and 10%, indicating high coherence and good alignment.

Introducing lenses to convert planar wave-fronts to spherical resulted in circular fringes. The fringe patterns can be seen in figure 6 and 7 respectively. According to section 2.7, we saw that he radii of fringes are given by

$$r_m = \sqrt{2Rm\lambda}, \qquad m = 0, 1, 2, \ldots$$

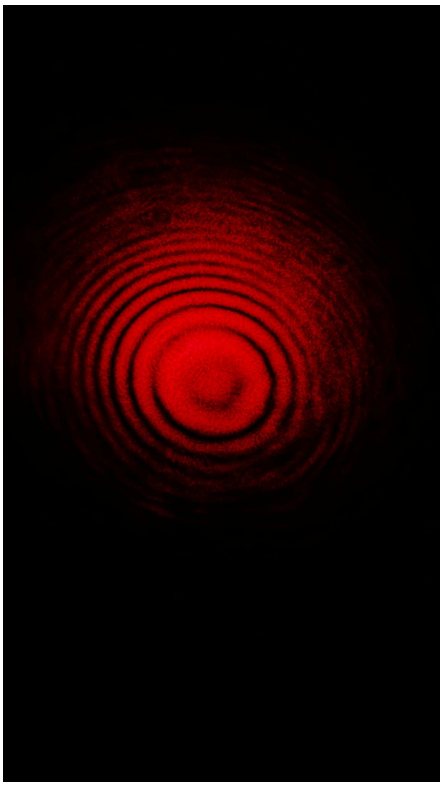


Figure 6: Circular fringe pattern observed due to interference of spherical and planar waves

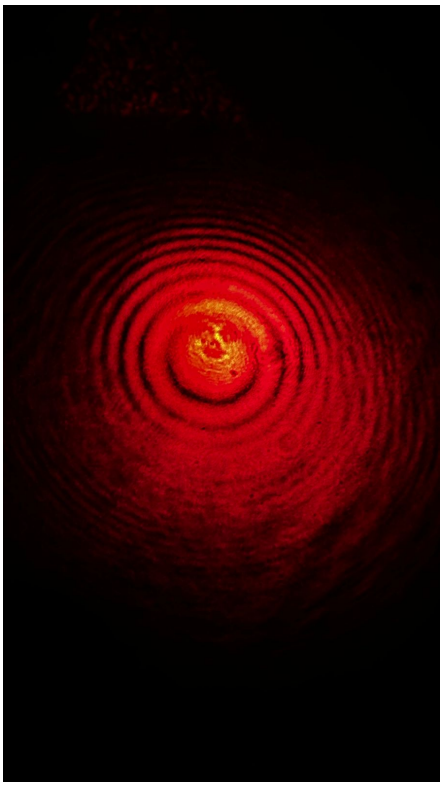


Figure 7: Circular fringe pattern observed due to interference of two spherical waves

For spherical and planar waves

$$r_m = \sqrt{\frac{m\,\lambda\,R_1 R_2}{|R_1 - R_2|}}, \qquad m = 0, 1, 2, \ldots$$

For spherical and spherical waves

Therfore, for our radii of curvature used, which are $R_1 = 10$cm and and $R_2 = 20$cm, we can see that the radius of fringes for spherical-spherical interference should grow $\sqrt{20/(20-10)} = \sqrt{2}$ times faster than the spherical-planar interference. We can see the from figure 6 and 7 that the radius of fringes for spherical-spherical interference indeed grows faster.

When orthogonal polarizations were introduced into each arm, the left arm became horizontally polarized and the right arm became vertically polarized but, the interference disappeared matching our prediction from section 2.8(see fig 8). Adding a polarizer on a rotating mount after recombination gradually restored fringe visibility, demonstrating the polarization-dependent suppression of interference(see figure 9, 10). Also, as predicted in section 3.2, the 45 degree polarizer outputs bright fringe in the center and the 135 degree outputs a dark fringe in the center but with similar patterns. This shows that polarization can effectively carry and reveal which-path information.

The interference output from the back-reflected beams can be seen in figure 11. It shows a bright fringe pattern. The observed pattern is bright, has high visibility due to large overlap

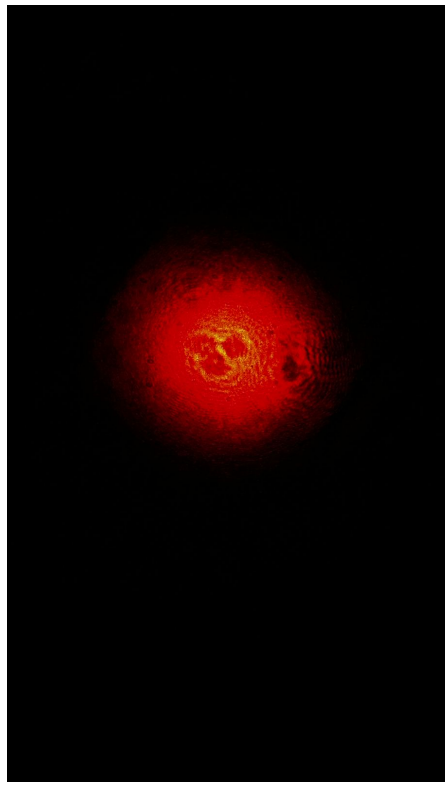


Figure 8: Interference of horizontally an vertically polarized light - no fringe patterns

and is also highly sensitive to path-length and vibration effects.

4.2 Michelson Interferometer

The Michelson interferometer produced circular fringe patterns when the arms were nearly equal in length(see figure 11). Slowly moving the piezo-actuated mirror caused fringes to sweep across the observation plane, demonstrating the sensitivity of the system to sub-wavelength changes in path length. Introducing deliberate angular misalignment reduced fringe contrast, consistent with the effect of spatial overlap and coherence. The maximum measured fringe visibility (see fig 12) was approximately 0.88, slightly lower than the Mach–Zehnder due to increased sensitivity to misalignment.

We tabulate the Max/Min voltage values and calculate the visibility for the three different mirror mismatch angles in Tables 2-4 and Plot the visibility vs mirror displacement to parabolic fits in figures 13 - 16. Then we fit them to parabolic fits given by

4.3 Parabolic Fits of Visibility and Coherence Length

The visibility of the interference fringes was measured as a function of the mirror displacement in the Mach–Zehnder interferometer for three different angular tilts: $\theta = 0.286°$, $0.143°$, and $0.429°$. Each dataset was fitted with a parabolic function of the form:

$$V(x) = ax^2 + bx + c,$$

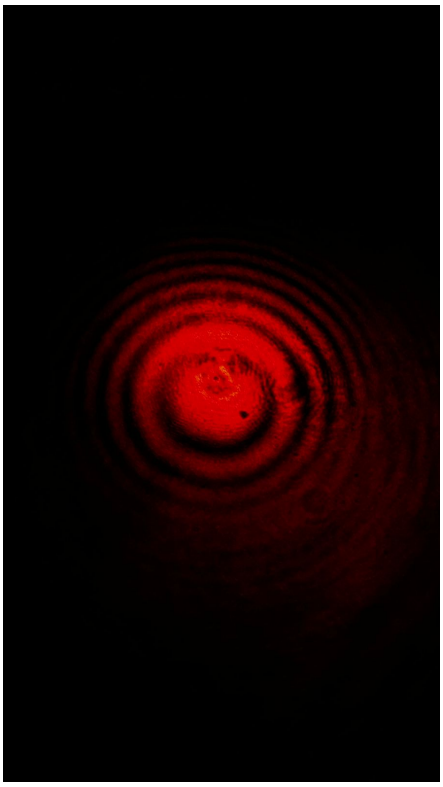


Figure 9: Interference of horizontally an vertically polarized light with polarizer at $\theta = 45°$ - Fringe pattern restored with bright central fringe

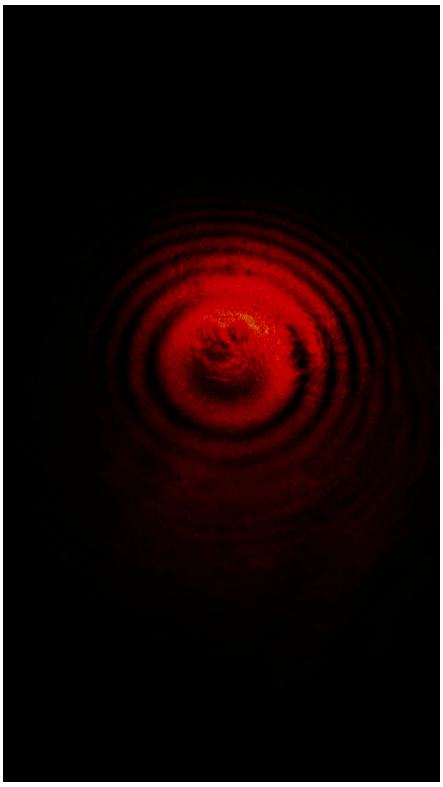


Figure 10: Interference of horizontally an vertically polarized light with polarizer at $\theta = 135°$ - Fringe pattern restored with dark central fringe

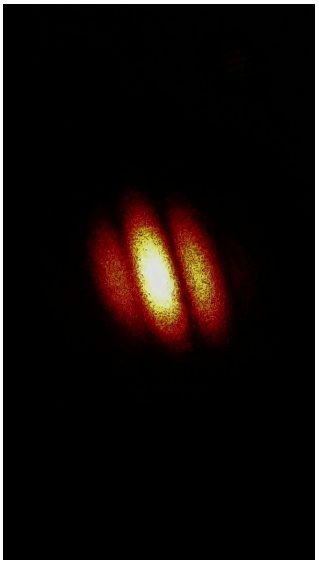


Figure 11: Interference of back-reflected rays outside BS1

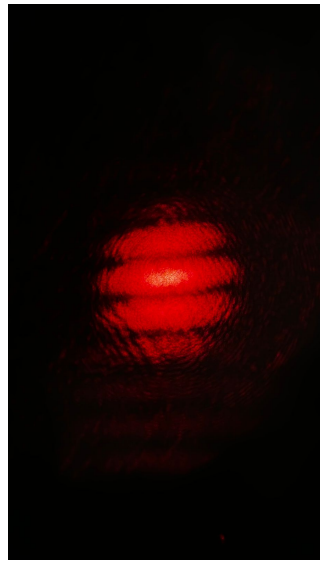


Figure 12: Interference pattern from Michelson Interferometer with both path lengths equal

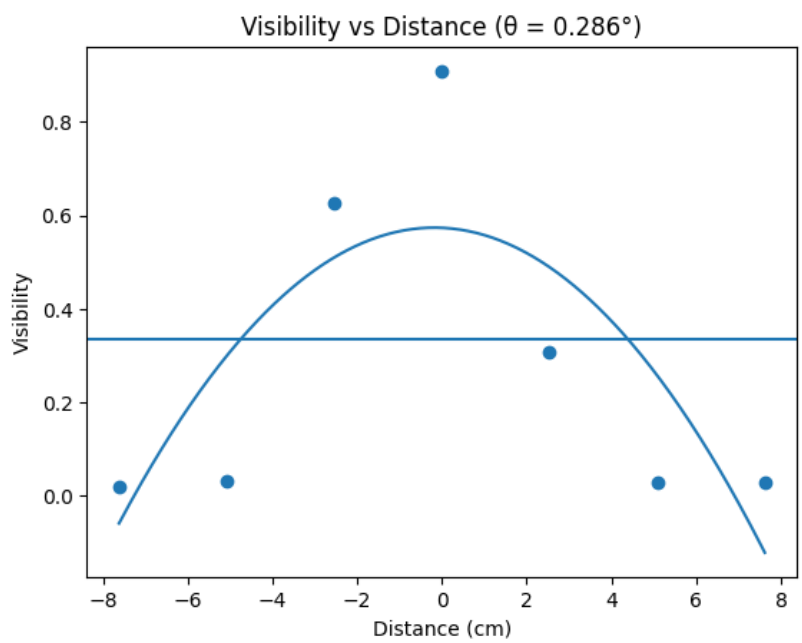


Figure 13: Measured Visibility vs mirror translation for mismatch of $\theta = 0.286°$

where V is the fringe visibility and x is the mirror displacement in centimeters. The fitted coefficients (a, b, c) for each dataset are summarized below.

Coherence Length Calculation The coherence length L_c for each dataset was estimated using the $1/e$ criterion, i.e., the mirror displacement at which the visibility drops to $V_{\max}/e$. Solving the quadratic equation

$$ax^2 + bx + (c - V_{\max}/e) = 0$$

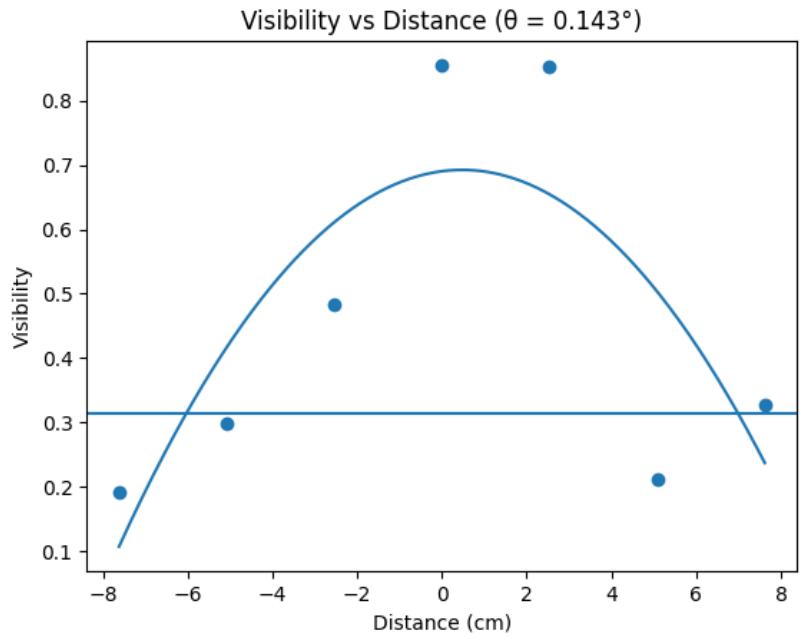


Figure 14: Measured Visibility vs mirror translation for mismatch of $\theta = 0.143°$

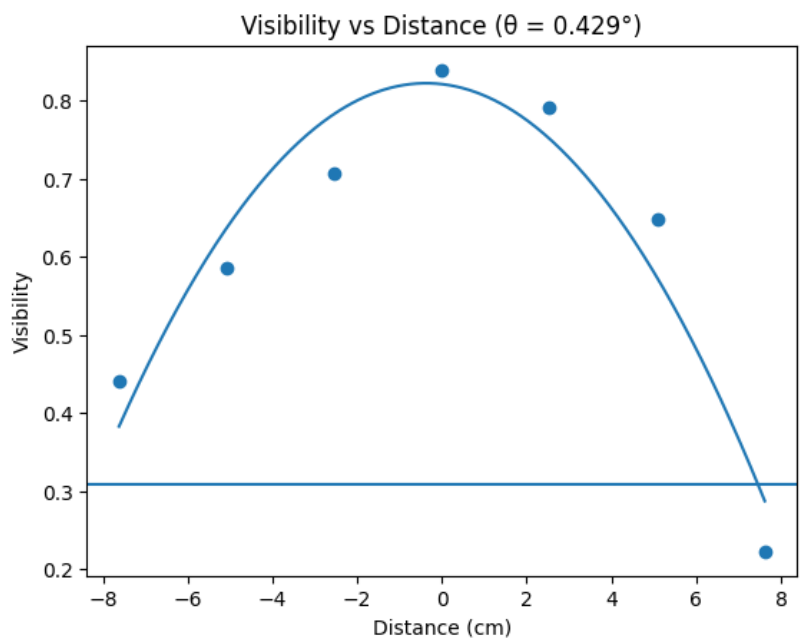


Figure 15: Measured Visibility vs mirror translation for mismatch of $\theta = 0.429°$

yields the two roots, corresponding to the mirror positions where the visibility falls to $1/e$ of the maximum. These roots are listed in Table 3.

Since the measured displacement corresponds to mirror translation, the actual optical path difference is twice the mirror displacement. Hence, the actual coherence length of the laser is taken as double the highest absolute value from the roots:

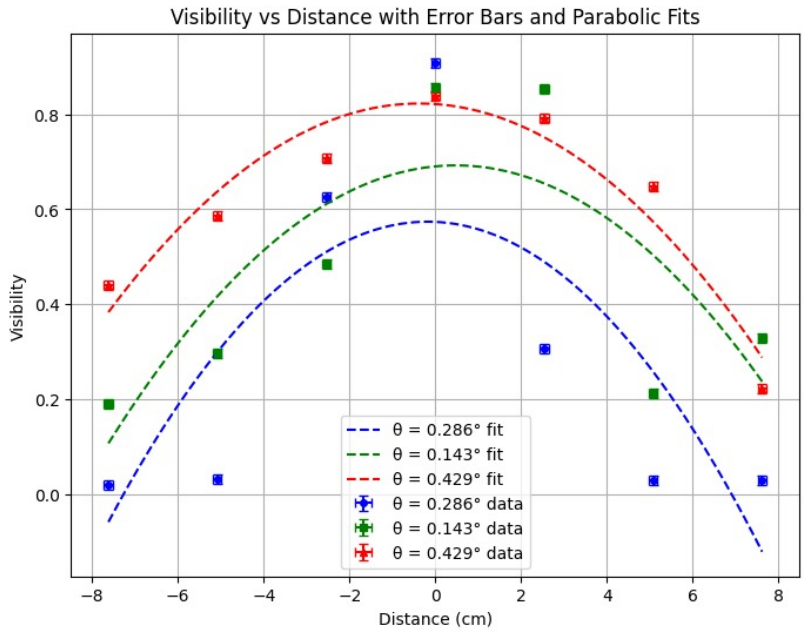


Figure 16: Realtive parabolic fits and dreived coherent lenghts

θ	a	b	c
0.286°	0.0241	-0.0163	0.912
0.143°	-0.0068	-0.0102	0.856
0.429°	-0.0324	-0.0185	0.840

Table 2: Fitted parabolic coefficients for the visibility as a function of mirror displacement.

θ	L_c (cm) from $1/e$ points
0.286°	0.86, 3.11
0.143°	-5.29, 5.29
0.429°	-2.74, 2.74

Table 3: Coherence lengths obtained from the $1/e$ visibility points.

$$L_{\text{coh}} = 2 \times \max(|x_{1/\text{e}}|).$$

This is why the largest root value is doubled—the light travels to the mirror and back, so the optical path difference changes by **twice the mirror movement**. Using this relation, the coherence lengths for the three datasets are:

$$\begin{aligned}
\theta = 0.286^\circ: &\quad L_{\text{coh}} = 2 \times 3.11 \text{ cm} \approx 6.22 \text{ cm}, \\
\theta = 0.143^\circ: &\quad L_{\text{coh}} = 2 \times 5.29 \text{ cm} \approx 10.58 \text{ cm}, \\
\theta = 0.429^\circ: &\quad L_{\text{coh}} = 2 \times 2.74 \text{ cm} \approx 5.48 \text{ cm}.
\end{aligned}$$

This approach accounts for the double path difference introduced by mirror translation,

which explains why the actual coherence length is twice the mirror displacement corresponding to the $1/e$ visibility point.

	Trial 1		Trial 2		Trial 3			Averages		
Distance (inch)	Max (V)	Min (V)	Max (V)	Min (V)	Max (V)	Min (V)	Distance (inch)	Avg Max (V)	Avg Min (V)	Visibility
-3	2.16	2.08	2.16	2.08	2.16	2.08	-3	2.16	2.08	0.0189
-2	2.18	2.04	2.16	2.06	2.20	2.04	-2	2.18	2.0467	0.0315
-1	3.49	0.80	3.51	0.78	3.49	0.84	-1	3.4967	0.8067	0.6251
0	4.05	0.17	4.11	0.11	4.11	0.13	0	2.8570	0.1367	0.9087
1	2.77	1.46	2.79	1.46	2.71	1.46	1	2.7567	1.4600	0.3075
2	2.12	2.00	2.12	2.00	2.12	2.00	2	2.1200	2.0000	0.0291
3	2.12	2.00	2.12	2.00	2.12	2.00	3	2.1200	2.0000	0.0291

Table 4: Measured maximum and minimum voltages at different mirror displacements for $\theta = 0.286°$. The fringe visibility is calculated using $V = (V_{\max} - V_{\min})/(V_{\max} + V_{\min})$.

	Trial 1		Trial 2		Trial 3			Averages		
Distance (inch)	Max (V)	Min (V)	Max (V)	Min (V)	Max (V)	Min (V)	Distance (inch)	Avg Max (V)	Avg Min (V)	Visibility
-3	2.54	1.60	2.40	1.72	2.48	1.72	-3	2.4733	1.6800	0.1910
-2	2.75	1.46	2.66	1.48	2.75	1.48	-2	2.7200	1.4733	0.2973
-1	3.07	1.10	3.13	1.16	3.17	1.00	-1	3.1233	1.0867	0.4838
0	3.95	0.25	3.77	0.41	3.95	0.25	0	3.8900	0.3033	0.8553
1	3.75	0.47	4.00	0.23	3.99	0.23	1	3.9133	0.3100	0.8532
2	2.50	1.66	2.60	1.64	2.58	1.70	2	2.5600	1.6667	0.2114
3	2.87	1.40	2.77	1.48	2.81	1.40	3	2.8167	1.4267	0.3276

Table 5: Measured maximum and minimum detector voltages for $\theta = 0.143°$. Fringe visibility calculated using $V = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$.

5 Conclusion

In this laboratory, classical optical interferometry was explored using both Mach–Zehnder and Michelson interferometers. The experiments successfully demonstrated the principles of interference, wavefront geometry, polarization effects, and coherence in a HeNe laser system.

In the Mach–Zehnder interferometer, we observed linear fringes arising from slight angular misalignment of planar wavefronts, and circular fringes resulting from the introduction of lenses to generate spherical wavefronts. The fringe radii and spacing closely matched theoretical predictions based on Gaussian beam optics and wavefront curvature. Additionally, polarization-based which-path information was shown to suppress interference, while introducing a polarizer after recombination restored fringe visibility in agreement with Malus' law

	Trial 1		Trial 2		Trial 3			Averages		
Distance (inch)	Max (V)	Min (V)	Max (V)	Min (V)	Max (V)	Min (V)	Distance (inch)	Avg Max (V)	Avg Min (V)	Visibility
-3	3.07	1.14	3.09	1.14	2.97	1.26	-3	3.0433	1.1800	0.4412
-2	3.33	0.94	3.43	0.84	3.37	0.86	-2	3.3767	0.8800	0.5865
-1	3.63	0.65	3.69	0.59	3.60	0.63	-1	3.6400	0.6233	0.7076
0	3.95	0.33	3.95	0.33	3.91	0.37	0	3.9367	0.3433	0.8396
1	3.45	0.80	4.05	0.17	3.89	0.35	1	3.7967	0.4400	0.7923
2	3.47	0.80	3.49	0.72	3.55	0.72	2	3.5033	0.7467	0.6486
3	2.54	1.64	2.56	1.64	2.58	1.60	3	2.5600	1.6267	0.2229

Table 6: Measured maximum and minimum detector voltages for $\theta = 0.429°$. Fringe visibility calculated using $V = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$.

and the concept of indistinguishability. This provided a classical analogue to the quantum eraser phenomenon, demonstrating the fundamental role of distinguishability in interference.

The Michelson interferometer exhibited high sensitivity to optical path differences, producing circular fringes that swept across the observation plane with mirror displacement. Measurement of fringe visibility as a function of mirror displacement allowed the extraction of the coherence length of the HeNe laser, confirming the relationship between temporal coherence and spectral linewidth. Varying angular misalignment highlighted the importance of spatial overlap for high-contrast interference.

Overall, the experiments confirmed that interference patterns are governed by phase differences, wavefront geometry, polarization, and coherence properties. The observations agreed quantitatively with theoretical models using paraxial approximation, Gaussian beam optics, ABCD matrix formalism, and polarization Jones calculus. These results reinforce the fundamental principles of classical interferometry and provide a foundation for understanding analogous quantum interference phenomena, such as single-photon Mach–Zehnder interference and quantum which-path experiments.

In conclusion, the laboratory successfully demonstrated that careful control of path lengths, beam alignment, wavefront curvature, and polarization is essential for observing and manipulating interference, providing both a practical and conceptual understanding of classical and quantum optical interferometry.

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